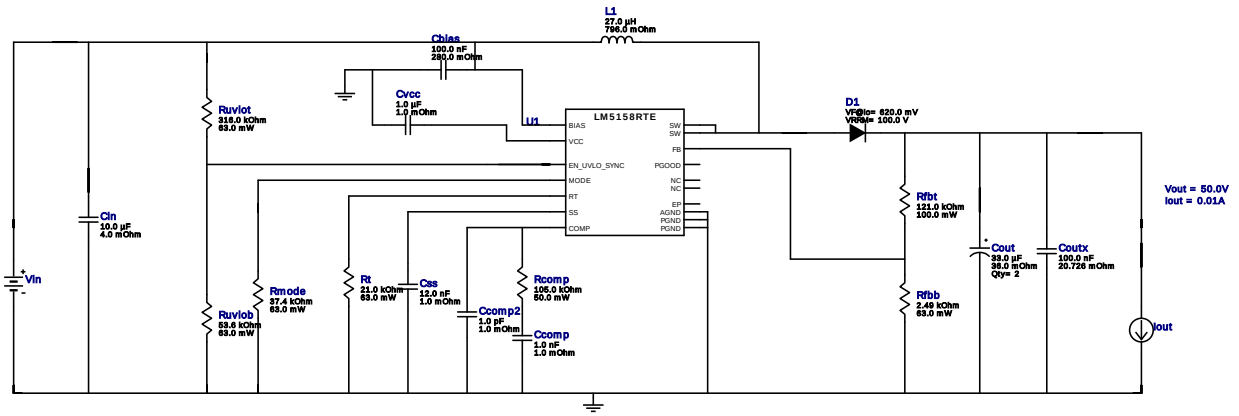
WEBENCH ® Design Report

VinMin = 10.0V
VinMax = 15.0V
Vout = 50.0V
Iout = 0.01A

Device = LM5158RTER
Topology = Boost
Created = 2026-03-08 14:11:38.085
BOM Cost = $5.62
BOM Count = 19
Total Pd = 0.04W

Design : 29 LM5158RTER
LM5158RTER 10V-15V to 50.00V @ 0.01A



Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cbias	AVX	08053C104KAT2A Series= X7R	Cap= 100.0 nF ESR= 280.0 mOhm VDC= 25.0 V IRMS= 0.0 A	1	$0.01	0805 7 mm²
Ccomp	MuRata	GRM155R60J102KA01D Series= X5R	Cap= 1.0 nF ESR= 1.0 mOhm VDC= 6.3 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Ccomp2	MuRata	GRM1555C1H1R0CA01D Series= C0G/NP0	Cap= 1.0 pF ESR= 1.0 mOhm VDC= 50.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Cin	MuRata	GRM21BR61E106MA73L Series= X5R	Cap= 10.0 uF ESR= 4.0 mOhm VDC= 25.0 V IRMS= 2.8 A	1	$0.04	0805 7 mm²
Cout	Panasonic	EEHZA1K330P Series= ZA	Cap= 33.0 uF ESR= 36.0 mOhm VDC= 80.0 V IRMS= 1.7 A	2	$1.34	SM_RADIAL_10BMM 160 mm²
Coutx	TDK	C2012X7R2A104K125AA Series= X7R	Cap= 100.0 nF ESR= 20.726 mOhm VDC= 100.0 V IRMS= 1.456 A	1	$0.03	0805 7 mm²
Css	MuRata	GRM155R71C123KA01D Series= X7R	Cap= 12.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0402 3 mm²
Cvcc	Taiyo Yuden	EMK107B7105KA-T Series= X7R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	$0.01	0603 5 mm²
D1	Vishay-Semiconductor	SS2PH10-M3	VF@Io= 620.0 mV VRRM= 100.0 V	1	$0.15	DO-220AA 14 mm²

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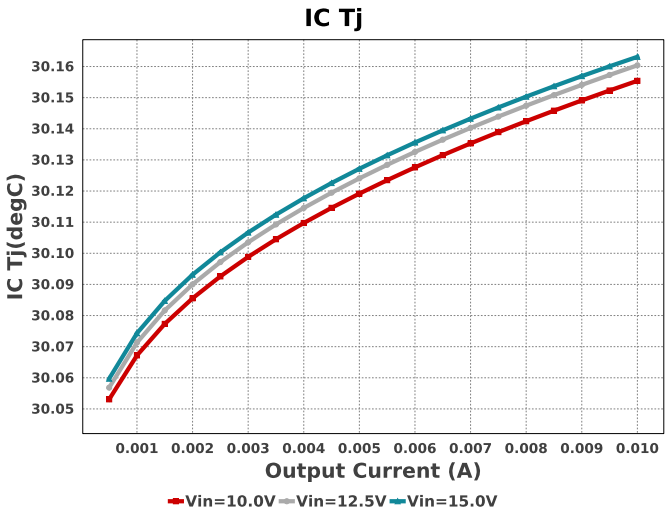
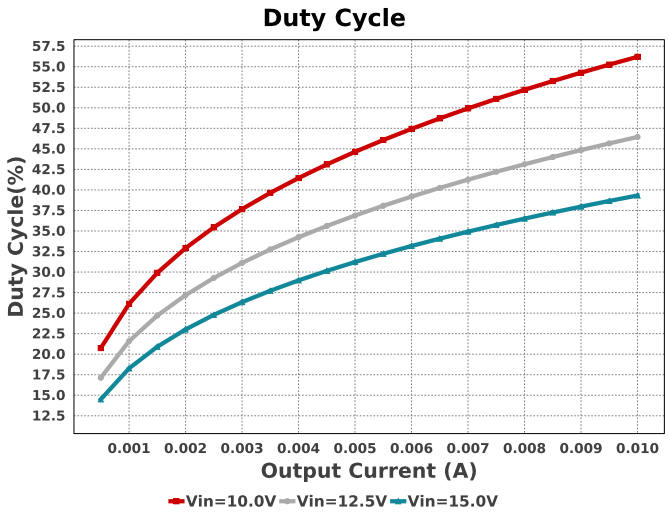
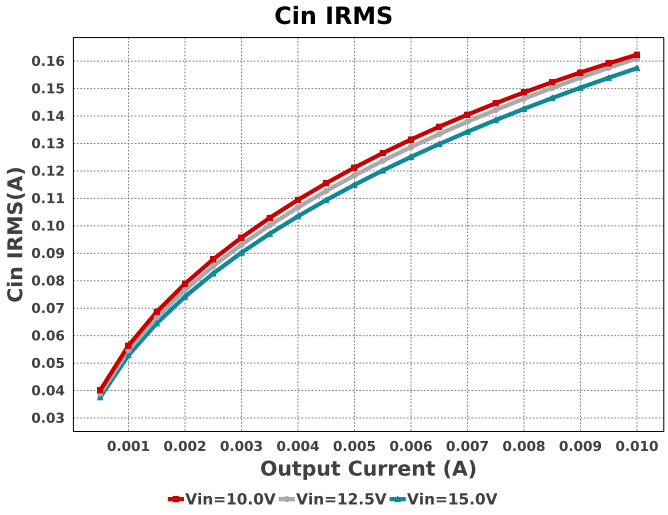
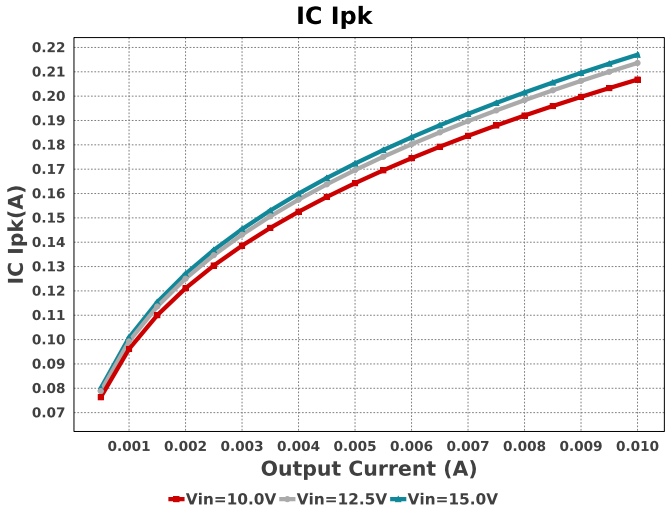
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
L1	Wurth Elektronik	744025270	L= 27.0 µH 796.0 mOhm	1	$0.77	WE-TPC_2828 14 mm²
Rcomp	Yageo	RC0201FR-07105KL Series= ?	Res= 105.0 kOhm Power= 50.0 mW Tolerance= 1.0%	1	$0.01	0201 2 mm²
Rfbb	Vishay-Dale	CRCW04022K49FKED Series= CRCW..e3	Res= 2.49 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rfbt	Vishay-Dale	CRCW0603121KFKEA Series= CRCW..e3	Res= 121.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	$0.01	0603 5 mm²
Rmode	Vishay-Dale	CRCW040237K4FKED Series= CRCW..e3	Res= 37.4 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Rt	Vishay-Dale	CRCW040221K0FKED Series= CRCW..e3	Res= 21.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Ruvlob	Vishay-Dale	CRCW040253K6FKED Series= CRCW..e3	Res= 53.6 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
Ruvlot	Vishay-Dale	CRCW0402316KFKED Series= CRCW..e3	Res= 316.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	$0.01	0402 3 mm²
U1	Texas Instruments	LM5158RTER	Switcher	1	$1.84	RTE0016K-IPC_A 16 mm²









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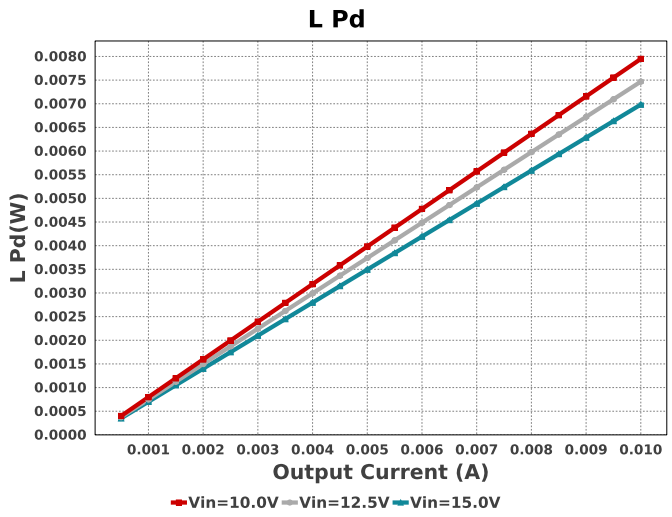

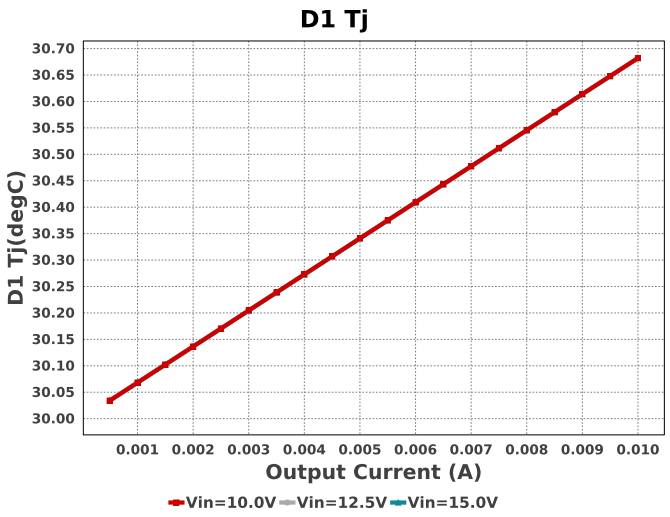

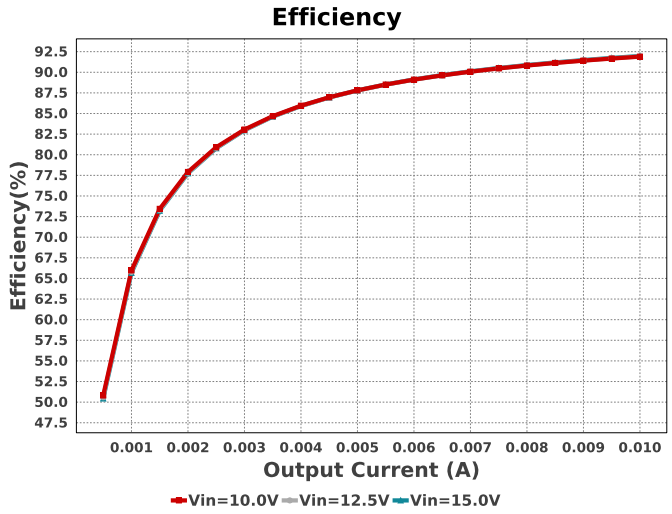

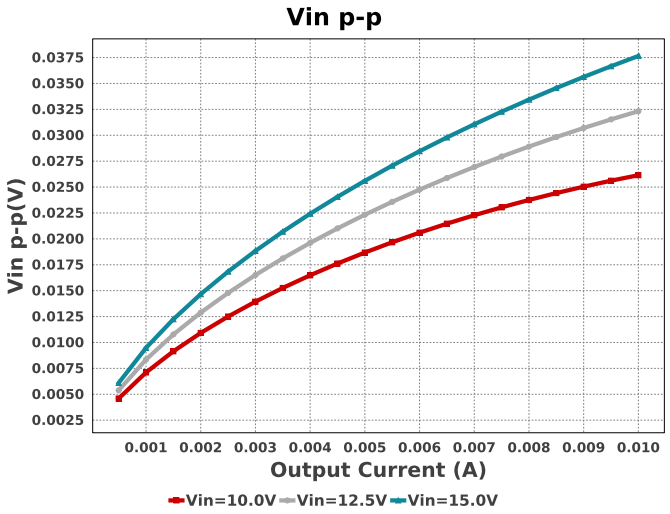

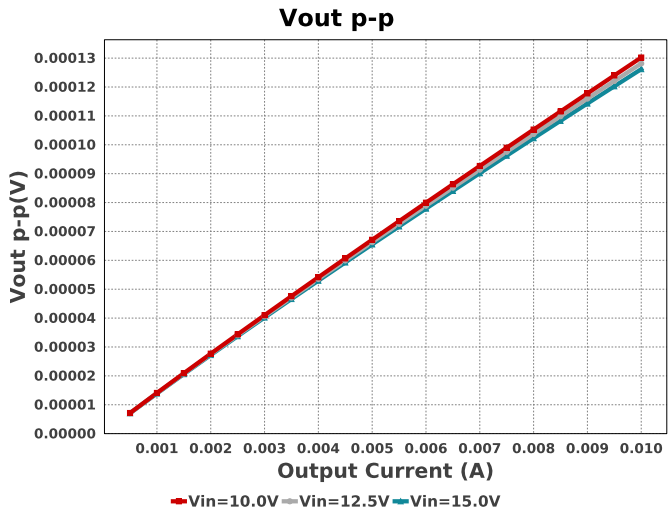

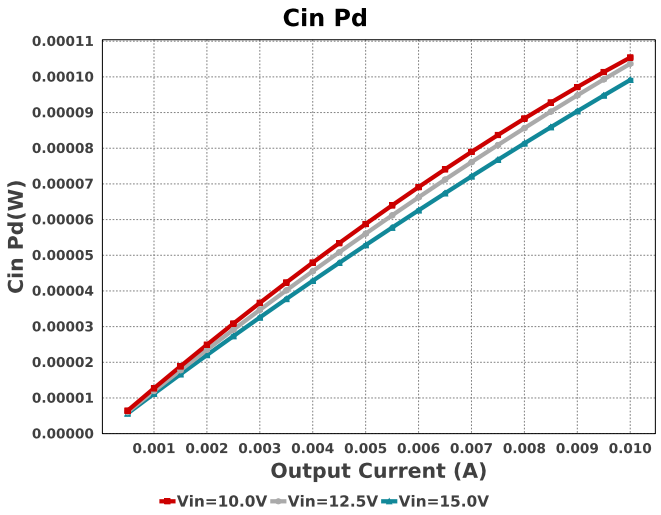

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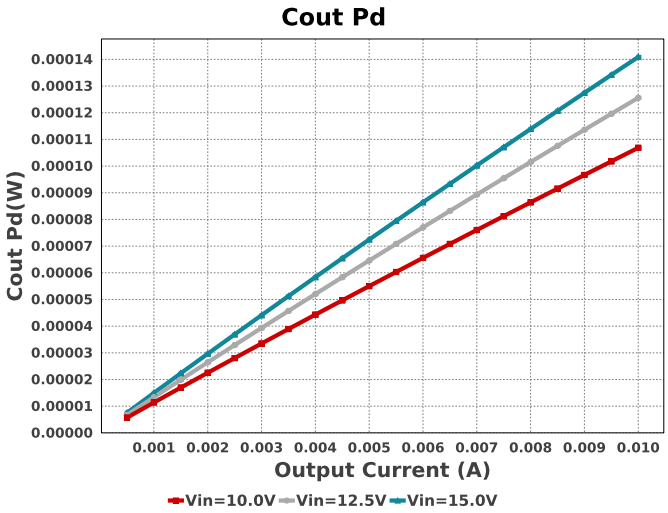

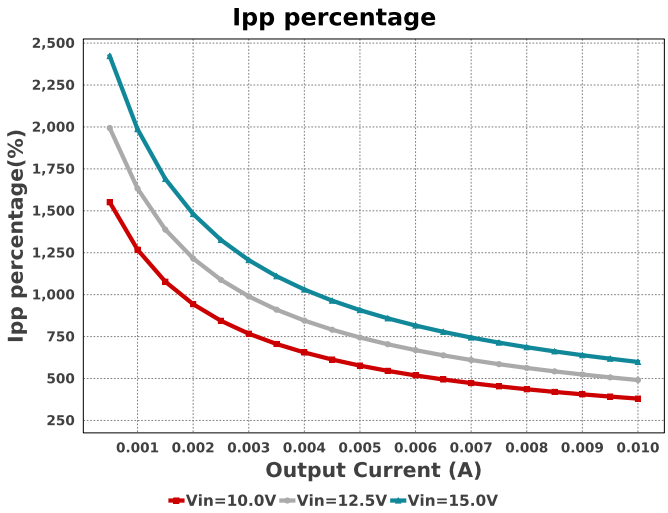

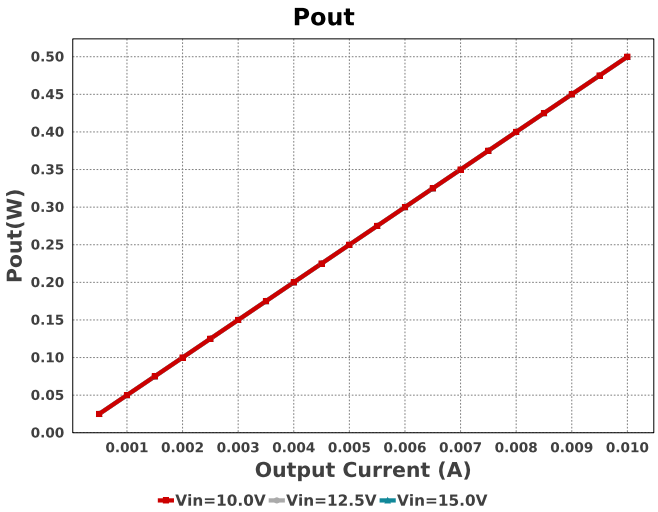

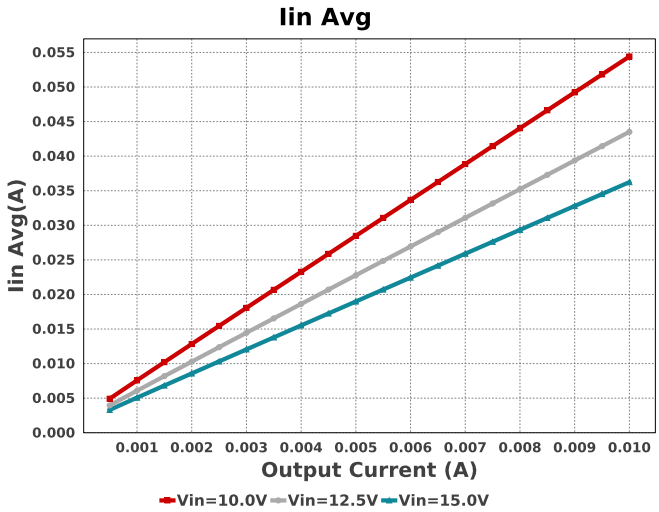

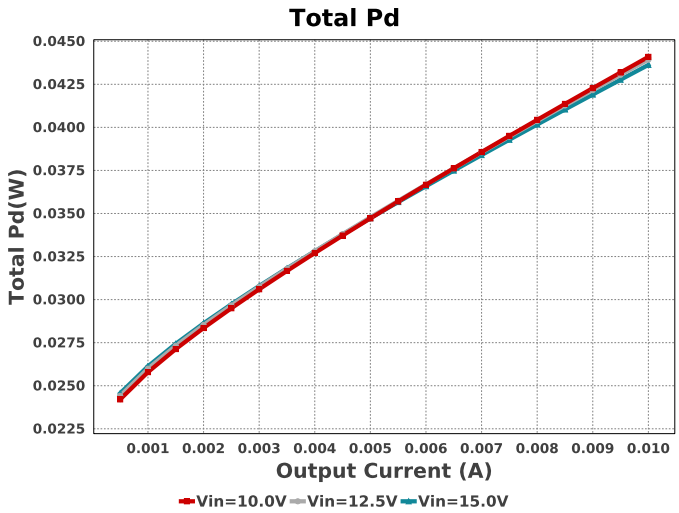

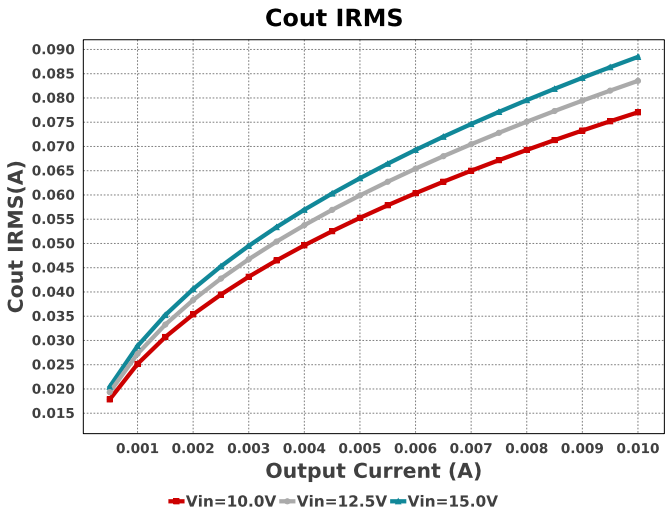

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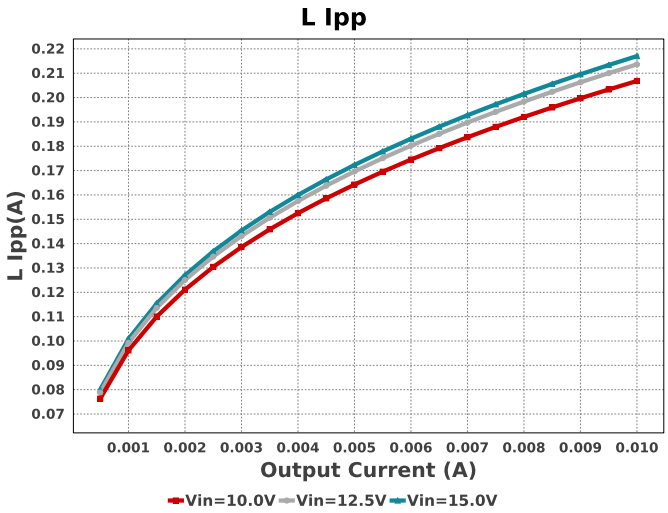

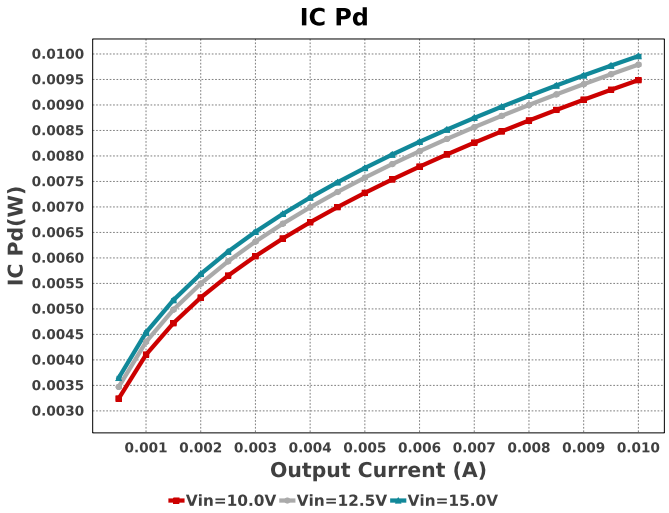

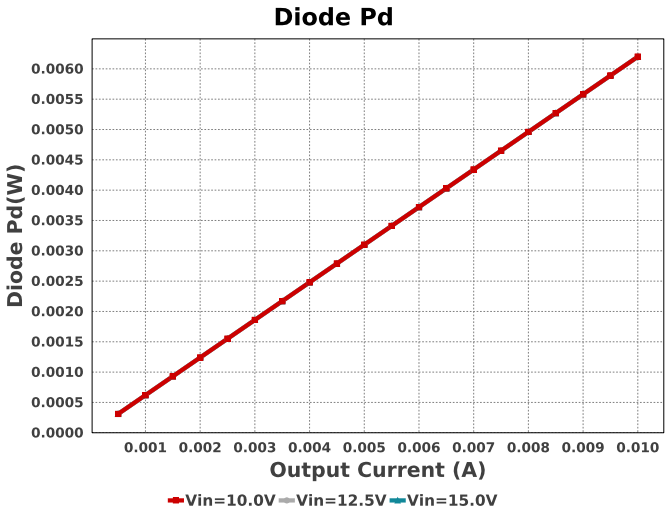

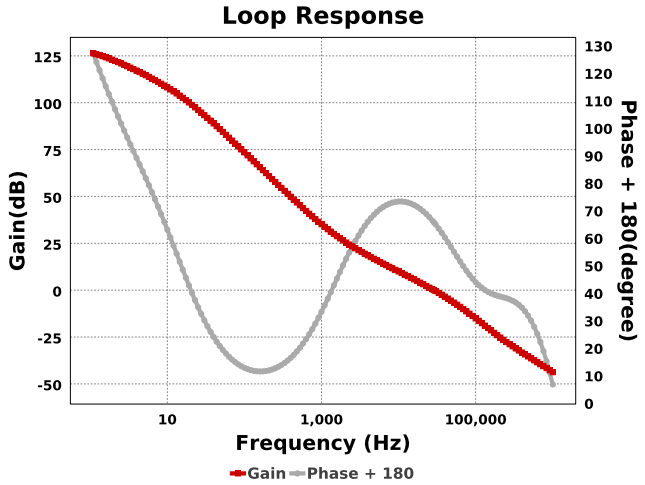

Operating Values

#	Name	Value	Category	Description
1.	BOM Count	19		Total Design BOM count
2.	Total BOM	$5.622		Total BOM Cost
3.	Cin IRMS	162.346 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	105.42 µW	Capacitor	Input capacitor power dissipation
5.	Cout IRMS	77.045 mA	Capacitor	Output capacitor RMS ripple current
6.	Cout Pd	106.85 µW	Capacitor	Output capacitor power dissipation
7.	D1 Tj	30.682 degC	Diode	D1 junction temperature
8.	Diode Pd	6.2 mW	Diode	Diode power dissipation
9.	IC Ipk	206.804 mA	IC	Peak switch current in IC
10.	IC Pd	9.487 mW	IC	IC power dissipation
11.	IC Tj	30.155 degC	IC	IC junction temperature
12.	IC Tolerance	10.0 mV	IC	IC Feedback Tolerance
13.	ICThetaJA Effective	16.38 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
14.	Iin Avg	54.409 mA	IC	Average input current
15.	Ipp percentage	380.09 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
16.	L Ipp	206.804 mA	Inductor	Peak-to-peak inductor ripple current
17.	L Pd	7.948 mW	Inductor	Inductor power dissipation
18.	Cin Pd	105.42 µW	Power	Input capacitor power dissipation
19.	Cout Pd	106.85 µW	Power	Output capacitor power dissipation
20.	Diode Pd	6.2 mW	Power	Diode power dissipation
21.	IC Pd	9.487 mW	Power	IC power dissipation
22.	L Pd	7.948 mW	Power	Inductor power dissipation
23.	Total Pd	44.093 mW	Power	Total Power Dissipation
24.	Cross Freq	20.197 kHz	System Information	Bode plot crossover frequency
25.	Duty Cycle	56.206 %	System Information	Duty cycle
26.	Efficiency	91.896 %	System Information	Steady state efficiency
27.	FootPrint	420.0 mm^2	System Information	Total Foot Print Area of BOM components

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#	Name	Value	Category	Description
28.	Frequency	1.007 MHz	System Information	Switching frequency
29.	Gain Marg	-42.993 dB	System Information	Bode Plot Gain Margin
30.	Iout	10.0 mA	System Information	Iout operating point
31.	Low Freq Gain	122.835 dB	System Information	Gain at 1Hz
32.	Mode	DCM	System Information	Conduction Mode
33.	Phase Marg	70.569 deg	System Information	Bode Plot Phase Margin
34.	Pout	500.0 mW	System Information	Total output power
35.	Vin	10.0 V	System Information	Vin operating point
36.	Vin p-p	26.145 mV	System Information	Input Source ripple voltage
37.	Vout	50.0 V	System Information	Operational Output Voltage
38.	Vout Actual	49.594 V	System Information	Vout Actual calculated based on selected voltage divider resistors
39.	Vout Tolerance	2.999 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
40.	Vout p-p	130.206 µV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	10.0 m	Maximum Output Current
SoftStart	1.0 ms	Soft Start Time (ms)
VinMax	15.0	Maximum input voltage
VinMin	10.0	Minimum input voltage
Vout	50.0	Output Voltage
base_pn	LM5158	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature
UserFsw	1000.0 k	Customer Selected Frequency

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WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of Cin and Cout, and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

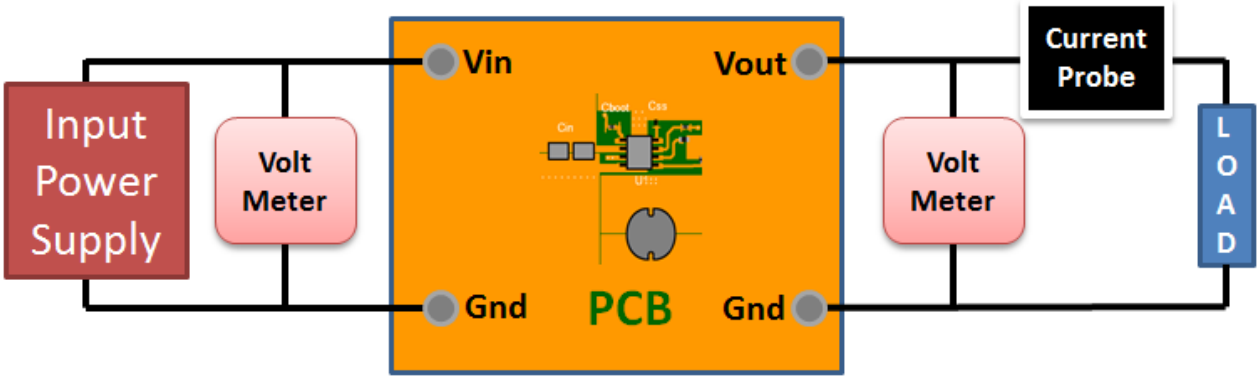
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab town to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 10.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to Vin and GND. Connect a digital volt meter and a load if needed to set the minimum Iout of the design from Vout and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between Vin and GND, a load is connected between Vout and GND and a current meter is connected in series between Vout and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 512A34938B868F72[v1]
2. **LM5158** Product Folder : http://www.ti.com/product/LM5158 : contains the data sheet and other resources.

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